

VQE For Quantum Chemistry

A physics-guided approach to setting up the VQE algorithm for application in quantum chemistry

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BU Quantum Seminar

Motivation: Quantum Chemistry

- General VQE solves for the lowest-energy eigenvector of an abstract Hamiltonian.
- Here, we will discuss adapting VQE to find the ground state of a many-electron molecular system.
- We will use principles from physics and chemistry to simplify the problem and set up the stage for the application of VQE.
- In chemistry, the ground state determines:
molecular geometry, chemical bonding, and the nature of chemical reactions.

Context: The Hydrogen Atom

- Write the Hamiltonian in a useful form and express as the “Time Independent Schrodinger Equation”:

$$\hat{\mathcal{H}}_{hydr}|\psi\rangle = E|\psi\rangle$$

- Perform “separation of variables” to get ODEs:

$$\psi(r, \theta, \phi) = R_{radial}(r)Y_{angular}(\theta, \phi) = R(r)\Theta(\theta)\Phi(\phi)$$

- Choose the eigenvector of the Time Independent Schrodinger Equation with the lowest eigenenergy: this is the ground state!

General Molecular Hamiltonian

$$\begin{aligned}\hat{H}_{mol} &= \hat{T}_N + \hat{T}_e + \hat{V}_{NN} + \hat{V}_{Ne} + \hat{V}_{ee} \\ &= -\sum_{a=1}^{N_N} \left(\frac{\hbar^2}{2M_a} \nabla_a^2 \right) - \sum_{i=1}^{N_e} \left(\frac{\hbar^2}{2m_e} \nabla_i^2 \right) + \frac{1}{2} \sum_A^{N_N} \sum_{B \neq A}^{N_N} \left(\frac{1}{4\pi\epsilon_0} \frac{(eZ_A)(eZ_B)}{|\vec{R}_A - \vec{R}_B|} \nabla_i^2 \right) \\ &\quad - \sum_{i=1}^{N_e} \sum_{A=1}^{N_N} \left(\frac{1}{4\pi\epsilon_0} \frac{(e)(eZ_A)}{|\vec{R}_A - \vec{r}_i|} \right) + \frac{1}{2} \sum_i^{N_N} \sum_{j \neq i}^{N_N} \left(\frac{1}{4\pi\epsilon_0} \frac{(e)(eZ_A)}{|\vec{R}_A - \vec{r}_i|} \right)\end{aligned}$$

Kinetic Energy of Wavefunction: $\hat{T}_x = -\frac{\hbar^2}{2m} \nabla_x^2$

Electric Potential Energy: $V(x) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{x}$

Molecular wavefunction: $\psi(\{\vec{r}_1, \vec{r}_2, \dots, \vec{r}_{N_e}\}, \{\vec{R}_1, \vec{R}_2, \dots, \vec{R}_{N_N}\})$

Molecular Hamiltonian

- Recall how we were able to separate the single wavefunction into a product of functions and solve for each of the functions individually. We can't do that here!

$$\frac{1}{2} \sum_A^{N_N} \sum_{B \neq A}^{N_N} \left(\frac{1}{4\pi\epsilon_0} \frac{(eZ_A)(eZ_B)}{|\vec{R}_A - \vec{R}_B|} \nabla_i^2 \right)$$

$$- \sum_{i=1}^{N_e} \sum_{A=1}^{N_N} \left(\frac{1}{4\pi\epsilon_0} \frac{(e)(eZ_A)}{|\vec{R}_A - \vec{r}_i|} \right)$$

$$\frac{1}{2} \sum_i^{N_e} \sum_{j \neq i}^{N_e} \left(\frac{1}{4\pi\epsilon_0} \frac{(e)(e)}{|\vec{r}_i - \vec{r}_j|} \right)$$

Mathematically coupled => Entangled!

List of Approximations

1. Born-Oppenheimer approximation:

Since $M_a \gg m_e$, we can treat nuclei as static, point charges.

2. Hartree approximation:

For the lack of a better approach, we will assume the electrons to be uncorrelated: $|\psi\rangle = |\phi_1\rangle \otimes |\phi_2\rangle \otimes \cdots \otimes |\phi_{N_e}\rangle$.

3. Hartree-Fock approximation:

True quantum chemistry requires anti-symmetrization (two electrons can't occupy the same quantum state). So, we will represent the system of electrons with a slater determinant:

$$|\psi\rangle = \begin{vmatrix} \phi_1(\vec{r}_1) & \phi_2(\vec{r}_1) \\ \phi_1(\vec{r}_2) & \phi_2(\vec{r}_2) \end{vmatrix} = \phi_1(\vec{r}_1)\phi_2(\vec{r}_2) - \phi_1(\vec{r}_2)\phi_2(\vec{r}_1)$$

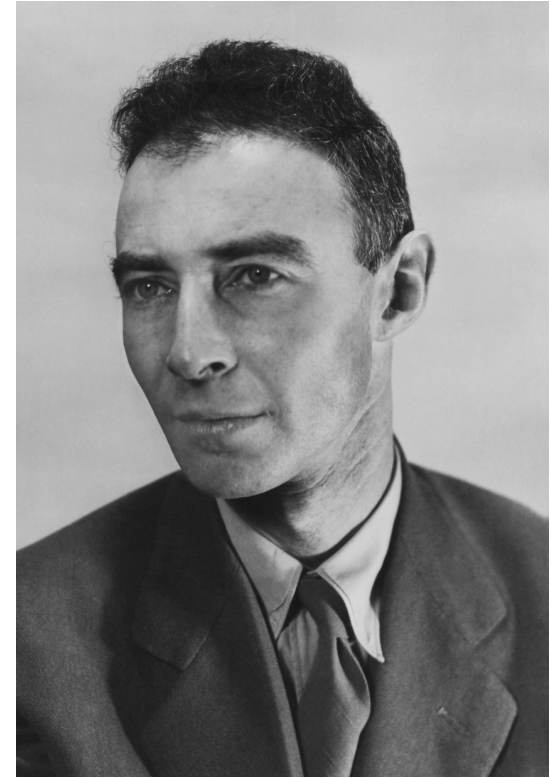
Born-Oppenheimer Approximation

- Because $M_a \gg m_e$:

$$T_e \approx \frac{p^2}{2m_e} \gg \frac{p^2}{2M_A} \approx T_N$$

- Separation of “timescales”:

$$\omega = \frac{T}{\hbar} \rightarrow \omega_e \gg \omega_N$$



Born-Oppenheimer Approximation

- The molecule's wavefunction separates!

$$V_{NN} = \frac{1}{2} \sum_A^{N_N} \sum_{B \neq A}^{N_N} \left(\frac{1}{4\pi\epsilon_0} \frac{(eZ_A)(eZ_B)}{|\vec{R}_A - \vec{R}_B|} \nabla_i^2 \right)$$

$$\begin{aligned} \psi(\{\vec{r}_1, \vec{r}_2, \dots, \vec{r}_{N_e}\}, \{\vec{R}_1, \vec{R}_2, \dots, \vec{R}_{N_N}\}) &= \\ &= \psi_e(\{\vec{r}_1, \vec{r}_2, \dots, \vec{r}_{N_e}\}, \{\vec{R}_1, \vec{R}_2, \dots, \vec{R}_{N_N}\}) \otimes \psi_N(\{\vec{R}_1, \vec{R}_2, \dots, \vec{R}_{N_N}\}) \end{aligned}$$

- We're able to only focus on the “electronic” part of the problem!

$$\begin{aligned} \hat{H}_{elec} &= \hat{T}_e + \hat{V}_{Ne} + \hat{V}_{ee} = \\ &= - \sum_{i=1}^{N_e} \left(\frac{\hbar^2}{2m_e} \nabla_i^2 \right) - \sum_{i=1}^{N_e} \sum_{A=1}^{N_N} \left(\frac{1}{4\pi\epsilon_0} \frac{(e)(eZ_A)}{|\vec{R}_A - \vec{r}_i|} \right) + \frac{1}{2} \sum_i^{N_e} \sum_{j \neq i}^{N_e} \left(\frac{1}{4\pi\epsilon_0} \frac{(e)(e)}{|\vec{r}_i - \vec{r}_j|} \right) \end{aligned}$$

Born-Oppenheimer Approximation

- The new TISE becomes:

$$\hat{H}_{elec} \psi_e(\dots) = E \psi_e(\dots)$$

- We treat the nuclei like classical point charges, calculating their energy as:

$$V_{NN} = \frac{1}{2} \sum_A^{N_N} \sum_{B \neq A}^{N_N} \left(\frac{1}{4\pi\epsilon_0} \frac{(eZ_A)(eZ_B)}{|\vec{R}_A - \vec{R}_B|} \nabla_i^2 \right)$$

Hartree Approximation

- For the lack of a better approach, we assume the electrons to be uncorrelated:

$$\psi_e(\{\vec{r}_1, \vec{r}_2, \dots, \vec{r}_{N_e}\}, \{\vec{R}_1, \vec{R}_2, \dots, \vec{R}_{N_N}\}) = \prod_{i=1}^{N_e} \psi_i(\vec{r}_i, \{\vec{R}\})$$

- Ignoring the electron correlation loses some information, but we will just use the result as an initial baseline for the ground state and then somehow recover the information that we lost.
(Sneak peak: using VQE)



Hartree Approximation

- Since the i -th electron only depends on its own position, $\vec{r}_i$:

$$\hat{V}_{ee}[\vec{r}_i, \vec{r}_j] \rightarrow \hat{V}_{ee}[\vec{r}_i]$$

- By reformulating the problem, we can rewrite $\hat{V}_{ee}[\vec{r}_i, \vec{r}_j]$ as:

$$\hat{V}_{ee}[\vec{r}_i] = \int_{-\infty}^{+\infty} \left(\frac{e^2}{4\pi\epsilon_0} \right) \frac{\sum_{j \neq i}^{N_e} \psi_j^*(\vec{r}_j) \psi_j(\vec{r}_j)}{|\vec{r}_i - \vec{r}_j|} d\vec{r}_j$$

A Different Perspective

- Again, recall what we have:

$$\hat{H}_{elec} \prod_{i=1}^{N_e} \psi_i(\vec{r}_i, \{\vec{R}\}) = (\hat{T}_e + \hat{V}_{Ne} + \hat{V}_{ee}[\vec{r}_i, \vec{r}_j]) \prod_{i=1}^{N_e} \psi_i(\vec{r}_i, \{\vec{R}\}) = E \prod_{i=1}^{N_e} \psi_i(\vec{r}_i, \{\vec{R}\})$$

- The $\hat{V}_{ee}[\vec{r}_i, \vec{r}_j]$ term still entangles the electrons, making an analytic solution nearly impossible (inseparable variables) and a numerical solution very hard.
- So... we should reformulate our question

A Different Perspective

- Right now, we're asking:

$$\hat{H}_{elec} \psi_e(\dots) = E \psi_e(\dots)$$

- What we ACTUALLY care about is minimizing E .
- We know from the variational principle:

$$E_{\lambda=0} \leq \frac{\langle \psi_e | H | \psi_e \rangle}{\langle \psi_e | \psi_e \rangle} \text{ for any } |\psi_e\rangle$$

- So, finding a $|\psi_e\rangle$ that minimizes $\langle \psi_e | H | \psi_e \rangle$ would answer the same question...

Lagrangian Multiplier Method

- We must turn the eigenequation into an optimization problem:

$$\hat{H}_{elec} \psi_e(\dots) = E \psi_e(\dots) \rightarrow \text{optimize}_{\psi_e} (\langle \psi_e | \hat{H} | \psi_e \rangle, \langle \psi_e | \psi_e \rangle)$$

- Set up a Lagrangian Multiplier expression:

$$\mathcal{F}[\psi_1, \dots, \psi_k, \dots, \psi_{N_e}] = \underbrace{\left\langle \prod_{i=1}^{N_e} \psi_i \left| \hat{H}_e \right| \prod_{i=1}^{N_e} \psi_i \right\rangle}_{\text{Quantity we'll minimize, by optimizing } \psi_k} - \sum_{k=1}^{N_e} \underbrace{\varepsilon_k (\langle \psi_k | \psi_k \rangle - 1)}_{\text{Constraint on } \psi_k:}$$

Quantity we'll minimize, by
optimizing ψ_k

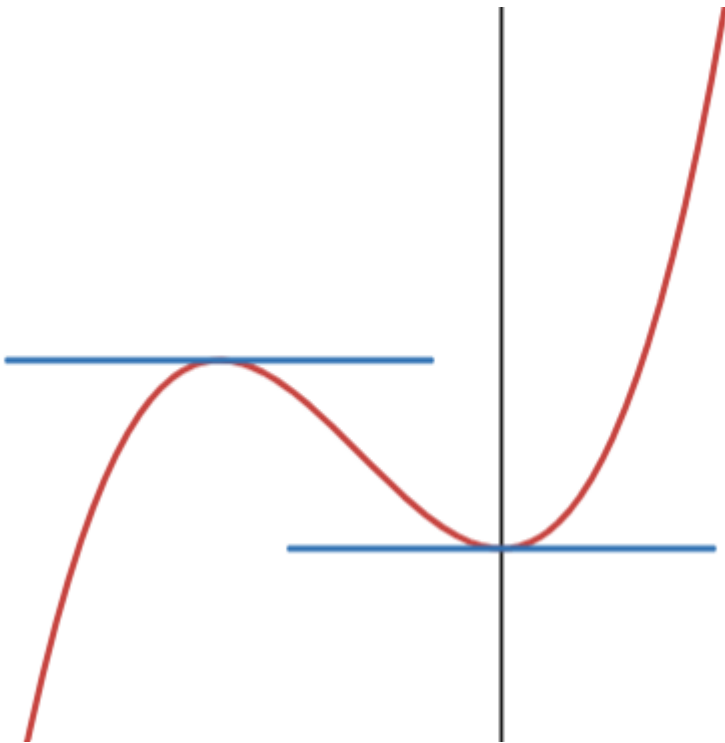
Constraint on ψ_k :

$$\langle \psi_k | \psi_k \rangle = 1$$

$$\langle \psi_k | \psi_k \rangle - 1 = 0$$

Lagrangian Multiplier Method

- Mathematically, we optimize functions using derivatives:



- To optimize the Lagrangian Multiplier expression (a functional), we will use functional derivatives:

$$\rightarrow \begin{cases} \frac{\delta \mathcal{F}}{\delta \psi_k^*} = 0 \\ \frac{\mathcal{F}}{\delta \epsilon_k^*} = 0 \end{cases} \quad \forall k \in [1, N_e]$$

Hartree Equation

- Solving the system of Lagrangian Multipliers produces:

$$\begin{cases} \frac{\delta \mathcal{F}}{\delta \psi_k^*} = 0 \\ \frac{\mathcal{F}}{\delta \varepsilon_k^*} = 0 \end{cases} \quad \forall k \in [1, N_e]$$


The Hartree Equation:

$$\hat{\mathcal{H}}[\{\psi_{j \neq k}\}] \psi_k(\vec{r}_k) = \left(-\frac{1}{2} \nabla_k^2 - \sum_{A=1}^{N_N} \frac{Z_A}{|\vec{r}_k - \vec{R}_A|} + \frac{1}{2} \int \frac{\sum_{j \neq k}^{N_e} (\psi_j^*(\vec{r}_j) \psi_j(\vec{r}_j))}{|\vec{r}_k - \vec{r}_j|} d\vec{r}_j \right) \psi_k(\vec{r}_k)$$
$$= \varepsilon_k \psi_k(\vec{r}_k)$$


- The Hartree operator represents the energy of the k -th electron:

$$\hat{\mathcal{H}}[\{\psi_{j \neq k}\}] \psi_k(\vec{r}_k) = \varepsilon_k \psi_k(\vec{r}_k)$$

Hartree Equation

- The “optimization problem” perspective allows us to focus on a single electron.

- Original:

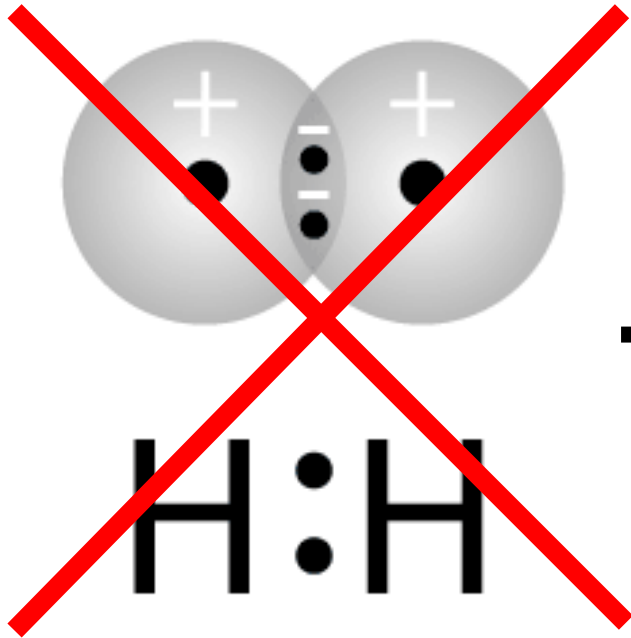
$$\hat{H}_{elec-k} = -\frac{1}{2}\nabla_k^2 - \sum_{A=1}^{N_N} \frac{Z_A}{|\vec{R}_A - \vec{r}_k|} + \frac{1}{2} \sum_{j \neq k}^{N_e} \frac{1}{|\vec{r}_k - \vec{r}_j|}$$

- Hartree Operator:

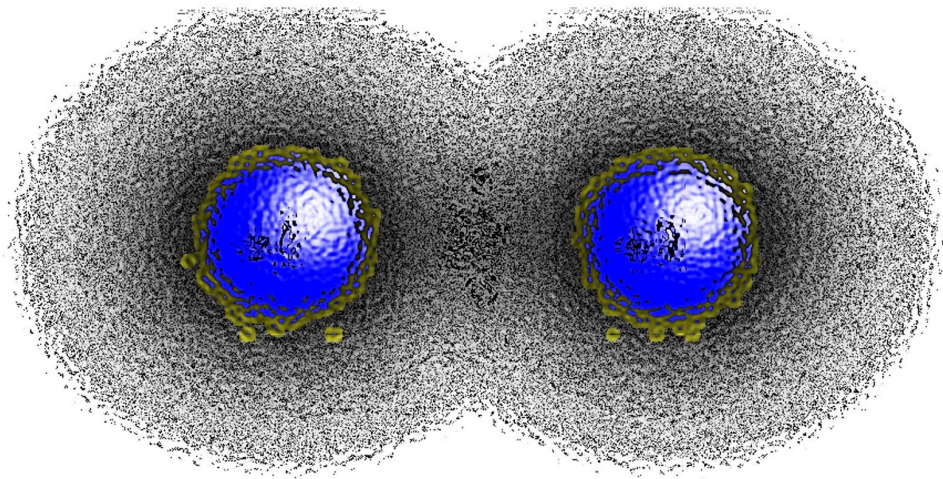
$$\hat{\mathcal{H}}[\{\psi_{j \neq k}\}] = -\frac{1}{2}\nabla_k^2 - \sum_{A=1}^{N_N} \frac{Z_A}{|\vec{r}_k - \vec{R}_A|} + \frac{1}{2} \int \frac{\sum_{j \neq k}^{N_e} (\psi_j^*(\vec{r}_j) \psi_j(\vec{r}_j))}{|\vec{r}_k - \vec{r}_j|} d\vec{r}_j$$

Identical Particles, Anyone?

- All electrons are identical to each other:



$$\psi_e(\vec{r}_1, \vec{r}_2) = \psi_1(\vec{r}_1) \otimes \psi_2(\vec{r}_2)$$



$$\psi_e(\vec{r}_1, \vec{r}_2) = -\psi_e(\vec{r}_2, \vec{r}_1)$$



Electron “anti-symmetry”

Hartree-Fock Approximation

- Pauli-Exclusion Principle:
No two electrons can occupy the same “wavefunction”

$$\psi_e(\vec{r}_1, \vec{r}_2) = \psi_1(\vec{r}_1)\psi_1(\vec{r}_2) - \psi_1(\vec{r}_2)\psi_1(\vec{r}_1) = 0$$

+

$$\psi_e(\vec{r}_1, \vec{r}_2) = -\psi_e(\vec{r}_2, \vec{r}_1)$$



$$\psi_e(\vec{r}_1, \vec{r}_2) = \psi_1(\vec{r}_1) \otimes \psi_2(\vec{r}_2) - \psi_1(\vec{r}_2) \otimes \psi_2(\vec{r}_1)$$

$$= \begin{vmatrix} \psi_1(\vec{r}_1) & \psi_2(\vec{r}_1) \\ \psi_1(\vec{r}_2) & \psi_2(\vec{r}_2) \end{vmatrix} = \psi_e(\vec{r}_1, \vec{r}_2) \quad \text{“Slater determinant”}$$

The Hartree-**Fock** Equation

- What we now have:

$$\psi_e(\vec{r}_1, \vec{r}_2) = \begin{vmatrix} \psi_1(\vec{r}_1) & \psi_2(\vec{r}_1) \\ \psi_1(\vec{r}_2) & \psi_2(\vec{r}_2) \end{vmatrix} \neq \psi_1(\vec{r}_1) \otimes \psi_2(\vec{r}_2)$$

- Solving the system of Lagrangian Multipliers again:

$$\begin{cases} \frac{\delta \mathcal{F}}{\delta \psi_k^*} = 0 \\ \frac{\mathcal{F}}{\delta \varepsilon_k^*} = 0 \end{cases} \quad \forall k \in [1, N_e]$$


The Hartree-**Fock** Equation:

$$\begin{aligned} \mathcal{F}[\{\psi_{j \neq k}\}] \psi_k(\vec{r}_k) &= \left(-\frac{1}{2} \nabla_k^2 - \sum_{A=1}^{N_N} \frac{Z_A}{|\vec{r}_k - \vec{R}_A|} + \int \frac{\sum_{j \neq k}^{N_e/2} (\psi_j^*(\vec{r}_j) \psi_j(\vec{r}_j))}{|\vec{r}_k - \vec{r}_j|} d\vec{r}_j - \frac{1}{2} \int \frac{\sum_{j \neq k}^{N_e/2} (\psi_j^*(\vec{r}_j) \text{ "—" } \psi_j(\vec{r}_j))}{|\vec{r}_k - \vec{r}_j|} d\vec{r}_j \right) \psi_k(\vec{r}_j) = \\ &= \varepsilon_k \psi_k(\vec{r}_k) \end{aligned}$$

The Hartree-Fock Equation

- Hartree-Fock operator:

$$\mathcal{F}[\{\psi_{j \neq k}\}] = \underbrace{-\frac{1}{2}\nabla_k^2 - \sum_{A=1}^{N_N} \frac{Z_A}{|\vec{r}_k - \vec{R}_A|}}_{\text{Single electron energy}} + \underbrace{\sum_{j \neq k}^{N_e/2} \int \frac{(\psi_j^*(\vec{r}_j)\psi_j(\vec{r}_j))}{|\vec{r}_k - \vec{r}_j|} d\vec{r}_j}_{\text{Coulomb (repulsion) energy}} - \underbrace{\frac{1}{2} \sum_{j \neq k}^{N_e/2} \int \frac{(\psi_j^*(\vec{r}_j) \text{ "—" } \psi_k(\vec{r}_j))}{|\vec{r}_k - \vec{r}_j|} d\vec{r}_j}_{\text{Exchange energy}}$$

Single electron energy:

$$\hat{h}_k = -\frac{1}{2}\nabla_k^2 - \sum_{A=1}^{N_N} \frac{Z_A}{|\vec{r}_k - \vec{R}_A|}$$

Coulomb (repulsion) energy:

$$\hat{J}_k[\{\psi_{j \neq k}\}] = \sum_{j \neq k}^{N_e/2} \int \frac{\psi_j^*(\vec{r}_j)\psi_j(\vec{r}_j)}{|\vec{r}_k - \vec{r}_j|} d\vec{r}_j$$

Exchange energy:

$$\hat{K}_k[\{\psi_{j \neq k}\}]\psi_k(\vec{r}_k) = \left[\frac{1}{2} \sum_{j \neq k}^{N_e/2} \int \frac{(\psi_j^*(\vec{r}_j)\psi_k(\vec{r}_j))}{|\vec{r}_k - \vec{r}_j|} d\vec{r}_j \right] \psi_j(\vec{r}_k)$$

- The Hartree-Fock operator represents the energy of the k -th orbital:

$$\mathcal{F}[\{\psi_{j \neq k}\}]\psi_k(\vec{r}_k) = \varepsilon_k \psi_k(\vec{r}_k)$$

The Exchange Interaction

- The exchange interaction is a result of electron anti-symmetry:

For distinguishable particles:

$$\psi_e(x_1, x_2) = \psi_1(x_1) \otimes \psi_2(x_2)$$

$$\begin{aligned}\langle \hat{X}_{\text{sep}} \rangle_{\text{dist}} &= \langle (x_1 - x_2)^2 \rangle \\ &= \langle x_1^2 \rangle + \langle x_2^2 \rangle - 2\langle x_1 \rangle \langle x_2 \rangle\end{aligned}$$

For indistinguishable particles:

$$\psi_e(x_1, x_2) = \psi_1(x_1) \otimes \psi_2(x_2)$$

$$\langle \hat{X}_{\text{sep}} \rangle_{\text{indist}} = \langle \hat{X}_{\text{sep}} \rangle_{\text{dist}} + 2 \underbrace{|\langle \psi_1 | \hat{X} | \psi_2 \rangle|^2}$$

Quantum “exchange” term:

$$\langle \psi_1 | \hat{X} | \psi_2 \rangle = \int_{-\infty}^{\infty} (\psi_1^*(x) \psi_2(x) \cdot x) dx$$

Recap of Hartree-Fock Method

$$\begin{aligned}\hat{H}_{mol} &= \hat{T}_N + \hat{T}_e + \hat{V}_{NN} + \hat{V}_{Ne} + \hat{V}_{ee} \\ &= -\sum_{a=1}^{N_N} \left(\frac{\hbar^2}{2M_a} \nabla_a^2 \right) - \sum_{i=1}^{N_e} \left(\frac{\hbar^2}{2m_e} \nabla_i^2 \right) + \frac{1}{2} \sum_A^{N_N} \sum_{B \neq A}^{N_N} \left(\frac{1}{4\pi\epsilon_0} \frac{(eZ_A)(eZ_B)}{|\vec{R}_A - \vec{R}_B|} \nabla_i^2 \right) - \sum_{i=1}^{N_e} \sum_{A=1}^{N_N} \left(\frac{1}{4\pi\epsilon_0} \frac{(e)(eZ_A)}{|\vec{R}_A - \vec{r}_i|} \right) + \frac{1}{2} \sum_i^{N_N} \sum_{j \neq i}^{N_N} \left(\frac{1}{4\pi\epsilon_0} \frac{(e)(eZ_A)}{|\vec{R}_A - \vec{r}_i|} \right)\end{aligned}$$

$$\hat{H}_{mol} \psi_{mol} = E_{mol} \psi_{mol}$$

1. Born-Oppenheimer approximation:

$$\hat{H}_{elec} \psi_e(\dots) = E \psi_e(\dots)$$

2. Hartree-Fock approximation:

$$\psi_e(\vec{r}_1, \vec{r}_2) = \begin{vmatrix} \psi_1(\vec{r}_1) & \psi_2(\vec{r}_1) \\ \psi_1(\vec{r}_2) & \psi_2(\vec{r}_2) \end{vmatrix}$$

Hartree-Fock Equation:

$$\mathcal{F}[\{\psi_{j \neq k}\}] \psi_k(\vec{r}_k) = \varepsilon_k \psi_k(\vec{r}_k)$$

Roothaan-Hall Approximation

- Now, we will convert the Hartree-Fock equation to a clean computational problem.
- We call the $\psi_k(\vec{r}_k)$ orbitals the “Molecular Orbitals”. These orbitals are our goal.
- To simplify their representation, we pick a basis set of M simpler, atom-centered “Atomic Orbitals”:

$$\{\chi_1, \dots, \chi_M\}$$

LCAO-MO Approximation

- We posit that **L**inear **C**ombinations of the **A**tomical **O**rbitals can accurately represent the **M**olecular **O**rbitals (**LCAO-MO**):

$$\psi_k(\vec{r}) = \sum_{a=1}^M (c_{ak} \chi_a(\vec{r}))$$

- Thus, we can represent the Hartree-Fock equation in the Atomic Orbital basis:

$$\mathcal{F}[\{\psi_{j \neq k}\}] \psi_k(\vec{r}_k) = \varepsilon_k \psi_k(\vec{r}_k) \rightarrow \langle \chi_b | \mathcal{F}[\{\psi_{j \neq k}\}] \sum_{a=1}^M (c_{ak} \chi_a(\vec{r})) = \langle \chi_b | \varepsilon_k \sum_{a=1}^M (c_{ak} \chi_a(\vec{r}))$$

Hartree-Fock Matrix Equation

- In the Atomic Orbital basis, the Hartree-Fock equation becomes a matrix equation:

$$\mathcal{F}_{M \times M} C_{M \times M} = C_{M \times M} E_{M \times M}$$

- Calculating $\psi_k(\vec{r}_k)$ for all k is a matter of diagonalizing the matrix equation to find $C_{M \times M}$.

- $C_{M \times M}$ describes the Molecular Orbitals:

$$C_{M \times M} = \begin{bmatrix} \downarrow & \downarrow & \downarrow \end{bmatrix}$$

Conceptually, expansion coefficients along the k -th column, correspond to the coefficients composing the k -th Molecular Orbital: $\psi_k(\vec{r}_k)$

Self-Consistent Fields Method

- The $C_{M \times M}$ and $F_{M \times M}$ matrices depend on each other:
a chicken and egg problem!
- We set up a loop to iterative construct the accurate system representation:
 1. Set $C_0 = [0]$, ignoring any electron-electron interactions.
 2. Construct the (single electron) energy landscape: $C_0 \rightarrow F_0, E_0$
 3. Diagonalize F_0 to find the spatial orbitals of the electrons: $F_0 \rightarrow C_1$
 4. Use C_1 to construct a more accurate energy landscape: $C_1 \rightarrow F_1, E_1$
 5. Repeat the loop until the results: $C_n \approx C_{n+1}$

The Ground State = Hartree-Fock State

- Once $C_{M \times M}$ converges, we use it to construct the optimal molecular orbitals:

$$C_{M \times M} = \left[\begin{array}{c} \downarrow \\ \downarrow \\ \downarrow \end{array} \right] \longrightarrow \psi_k(\vec{r}) = \sum_{a=1}^M (c_{ak} \mathcal{X}_a(\vec{r})) \longrightarrow \begin{array}{l} \text{MO:} \\ \{\psi_1(\vec{r}), \dots, \psi_M(\vec{r})\} \end{array}$$

- We construct the Hartree-Fock ground state by plugging the optimized molecular orbitals into the Slater determinant:

$$\psi_{e-GR} = \det \left(\{ \psi_1(\vec{r}), \dots, \psi_{\frac{M}{2}}(\vec{r}) \} \right)$$

What's Next?

- Excerpt from **Ideas of Quantum Chemistry**:

“The correlation of the motions of the electrons results in an energy gain called correlation energy. According to the Hartree–Fock (HF) method, the F₂ molecule does not exist at all, whereas in reality, it exists with bonding energy equal to 38kcal/mol. The mean field method provides ca. 99% of the total energy of the system.”

- Post Hartree-Fock methods:

1. Møller–Plesset Perturbation Theory
2. Configuration Interaction
3. VQE

Putting $\hat{H}_{mol}$ Onto a Quantum Computer

- The representation of a Hamiltonian in “second-quantized form” is:

$$\hat{H}_{mol} = I^{\otimes 2M} \underbrace{(V_{NN})}_{\text{Nuclear Repulsion Energy}} + \sum_{p,q} \underbrace{(h_{pq} \hat{\alpha}_p^\dagger \hat{\alpha}_q)}_{\text{Single Electron (Occupation \& Transition) Energy}} + \frac{1}{2} \sum_{p,q,r,s} \underbrace{(h_{pqrs} \hat{\alpha}_p^\dagger \hat{\alpha}_q^\dagger \hat{\alpha}_s \hat{\alpha}_r)}_{\text{Electron Pair (Occupation \& Transition) Energy}}$$

Nuclear Repulsion Energy:

$$V_{NN} = \frac{1}{2} \sum_A^{N_N} \sum_{B \neq A}^{N_N} \left(\frac{1}{4\pi\epsilon_0} \frac{(eZ_A)(eZ_B)}{|\vec{R}_A - \vec{R}_B|} \nabla_i^2 \right)$$

Electron Pair (Occupation & Transition) Energy:

$$h_{pqrs} = \int \int \psi_p^*(\vec{r}_1) \psi_q^*(\vec{r}_2) \left(\frac{1}{|\vec{r}_1 - \vec{r}_2|} \right) \psi_r(\vec{r}_2) \psi_s(\vec{r}_1) d\vec{r}_1 d\vec{r}_2$$

Single Electron (Occupation & Transition) Energy:

$$h_{pq} = \langle \hat{T}_e + \hat{V}_{Ne} \rangle = \int \psi_p^*(\vec{r}) \left(-\frac{\nabla^2}{2} - \sum_{A=1}^{N_n} \frac{Z_A}{|\vec{r} - \vec{R}_A|} \right) \psi_q(\vec{r}) d\vec{r}$$

Occupation Number Representation

- Recall assigning electrons to spatial orbitals:

$$\psi_e(\vec{r}_1, \vec{r}_2) = \psi_1(\vec{r}_1) \otimes \psi_2(\vec{r}_2) - \psi_1(\vec{r}_2) \otimes \psi_2(\vec{r}_1)$$

- Suppose there are 4 orthogonal orbitals and only 2 electrons:

$$\psi_e(\vec{r}_1, \vec{r}_2) \rightarrow \{\psi_1, \psi_2, \psi_3, \psi_4\}$$

- We may assign the 2 electrons to the orbitals as:

$$\vec{s} = \langle a, b, c, d \rangle \in \{0, 1\}^{2M} \longrightarrow \psi_e = [\psi_1]_{e1}^a \otimes [\psi_2]_{e2}^b \otimes [\psi_3]_{e3}^c \otimes [\psi_4]_{e4}^d$$

$$\text{S.T. : } a + b + c + d = 2$$

Transitioning to Second Quantization

- For a physical representation of the system, we must satisfy the Pauli-Exclusion Principle and electron anti-symmetrization:

$$\psi_e = [\psi_1]_{e1}^a \otimes [\psi_2]_{e2}^b \otimes [\psi_3]_{e3}^c \otimes [\psi_4]_{e4}^d \quad [\psi_1]_{e1}^a \otimes [\psi_2]_{e2}^b \otimes [\psi_3]_{e3}^c \otimes [\psi_4]_{e4}^d =$$

$$\text{S.T. : } a + b + c + d = 2 \quad = -[\psi_2]_{e1}^a \otimes [\psi_1]_{e2}^b \otimes [\psi_3]_{e3}^c \otimes [\psi_4]_{e4}^d$$

- We use an “occupation-number” representation of the system and assign electrons with “creation” and “annihilation” operators:

$$\psi_e(r_1, r_2) = \prod_{k=1}^{2M=4} |\{0, 1\}\rangle_{e_k} \quad \hat{\alpha}_k^\dagger, \hat{\alpha}_k$$

Second Quantization 101

- The “Creation” operator:

$$\hat{\alpha}_k^\dagger |0\rangle_k = 1 |1\rangle_k, \quad \hat{\alpha}_k^\dagger |1\rangle_k = 0$$

- The “Annihilation” operator:

$$\hat{\alpha}_k |0\rangle_k = 0, \quad \hat{\alpha}_k |1\rangle_k = 1 |0\rangle_k$$

I.e., we associate the k -th qubit with the k -th orbital, and use the qubit’s two orthogonal states to represent an empty orbital $|0\rangle$ or one that is occupied by an electron $|1\rangle$.

- The “Number” Operator:

$$(\hat{\alpha}_k^\dagger \hat{\alpha}_k) |1\rangle = \hat{\alpha}_k^\dagger (1 |0\rangle) = 1 |1\rangle$$

$$(\hat{\alpha}_k^\dagger \hat{\alpha}_k) |0\rangle = \hat{\alpha}_k^\dagger (0 |0\rangle) = 0 |0\rangle$$

Canonical Anti-Commutation Relations

- 2 simple rules generate all of the properties above.
- Notably, these 2 rules naturally encode anti-symmetrization:

$$\{\hat{\alpha}_j, \hat{\alpha}_k^\dagger\} = \delta_{jk} \hat{I}$$



$$\begin{cases} \delta_{j \neq k} \hat{I} = 0 = \{\hat{\alpha}_j, \hat{\alpha}_k^\dagger\} = \hat{\alpha}_j \hat{\alpha}_k^\dagger + \hat{\alpha}_k^\dagger \hat{\alpha}_j \\ \delta_{j \neq k}^\dagger \hat{I}^\dagger = 0 = \{\hat{\alpha}_j, \hat{\alpha}_k^\dagger\}^\dagger = \hat{\alpha}_j^\dagger \hat{\alpha}_k + \hat{\alpha}_k \hat{\alpha}_j^\dagger \end{cases}$$



$$\begin{cases} \hat{\alpha}_j \hat{\alpha}_k^\dagger = -\hat{\alpha}_k^\dagger \hat{\alpha}_j \\ \hat{\alpha}_j^\dagger \hat{\alpha}_k = -\hat{\alpha}_k \hat{\alpha}_j^\dagger \end{cases}$$

$$\{\hat{\alpha}_j^\dagger, \hat{\alpha}_k^\dagger\} = \{\hat{\alpha}_j, \hat{\alpha}_k\} = 0$$



$$\begin{cases} \{\hat{\alpha}_j^\dagger, \hat{\alpha}_k^\dagger\} = \hat{\alpha}_j^\dagger \hat{\alpha}_k^\dagger + \hat{\alpha}_k^\dagger \hat{\alpha}_j^\dagger = 0 \\ \{\hat{\alpha}_j, \hat{\alpha}_k\} = \hat{\alpha}_j \hat{\alpha}_k + \hat{\alpha}_k \hat{\alpha}_j = 0 \end{cases}$$



$$\begin{cases} \hat{\alpha}_j^\dagger \hat{\alpha}_k^\dagger = -\hat{\alpha}_k^\dagger \hat{\alpha}_j^\dagger \\ \hat{\alpha}_j \hat{\alpha}_k = -\hat{\alpha}_k \hat{\alpha}_j \end{cases}$$

Canonical Anti-Commutation Relations

- Returning back to our “4 orbitals and 2 electrons” problem:

$$\psi_e = [\psi_1]_{e1}^1 \otimes [\psi_2]_{e2}^1 \otimes [\psi_3]_{e3}^0 \otimes [\psi_4]_{e4}^0$$



$$\psi_{empty} |0000\rangle \rightarrow \hat{\alpha}_1^\dagger \hat{\alpha}_2^\dagger \hat{I} \hat{I} |0000\rangle = |1100\rangle$$

$$\psi_{empty} |0000\rangle \rightarrow -\hat{\alpha}_2^\dagger \hat{\alpha}_1^\dagger \hat{I} \hat{I} |0000\rangle = -|1100\rangle$$

$$\hat{\alpha}_1^\dagger \hat{\alpha}_2^\dagger = -\hat{\alpha}_2^\dagger \hat{\alpha}_1^\dagger$$

Putting $\hat{H}_{mol}$ Onto a Quantum Computer

- The representation of a Hamiltonian in “second-quantized form” is:

$$\hat{H}_{mol} = I^{\otimes 2M} \underbrace{(V_{NN})}_{\text{Nuclear Repulsion Energy}} + \sum_{p,q} \underbrace{(h_{pq} \hat{\alpha}_p^\dagger \hat{\alpha}_q)}_{\text{Single Electron (Occupation \& Transition) Energy}} + \frac{1}{2} \sum_{p,q,r,s} \underbrace{(h_{pqrs} \hat{\alpha}_p^\dagger \hat{\alpha}_q^\dagger \hat{\alpha}_s \hat{\alpha}_r)}_{\text{Electron Pair (Occupation \& Transition) Energy}}$$

Nuclear Repulsion Energy:

$$V_{NN} = \frac{1}{2} \sum_A^{N_N} \sum_{B \neq A}^{N_N} \left(\frac{1}{4\pi\epsilon_0} \frac{(eZ_A)(eZ_B)}{|\vec{R}_A - \vec{R}_B|} \nabla_i^2 \right)$$

Electron Pair (Occupation & Transition) Energy:

$$h_{pqrs} = \int \int \psi_p^*(\vec{r}_1) \psi_q^*(\vec{r}_2) \left(\frac{1}{|\vec{r}_1 - \vec{r}_2|} \right) \psi_r(\vec{r}_2) \psi_s(\vec{r}_1) d\vec{r}_1 d\vec{r}_2$$

Single Electron (Occupation & Transition) Energy:

$$h_{pq} = \langle \hat{T}_e + \hat{V}_{Ne} \rangle = \int \psi_p^*(\vec{r}) \left(-\frac{\nabla^2}{2} - \sum_{A=1}^{N_n} \frac{Z_A}{|\vec{r} - \vec{R}_A|} \right) \psi_q(\vec{r}) d\vec{r}$$

Jordan Wigner Mapping

- In our qubit representation of electron occupation, it is natural that:

$$\begin{cases} \hat{\alpha}_k^\dagger |0\rangle_k = 1 |1\rangle_k \\ \hat{\alpha}_k^\dagger |1\rangle_k = 0 \end{cases} \rightarrow \hat{\alpha}_k^\dagger = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}_k \longrightarrow \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}_k = \frac{1}{2} (\hat{X}_k - i\hat{Y}_k)$$

$$\begin{cases} \hat{\alpha}_k |0\rangle_k = 0 \\ \hat{\alpha}_k |1\rangle_k = 1 |0\rangle_k \end{cases} \rightarrow \hat{\alpha}_k = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}_k \longrightarrow \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}_k = \frac{1}{2} (\hat{X}_k + i\hat{Y}_k)$$

Jordan Wigner Mapping

- Problem with anti-symmetrization:

$$[\hat{A}_j, \hat{B}_k] = 0 \rightarrow \frac{1}{2} (\hat{X}_1 - i\hat{Y}_1) \frac{1}{2} (\hat{X}_2 - i\hat{Y}_2) = + \frac{1}{2} (\hat{X}_2 - i\hat{Y}_2) \frac{1}{2} (\hat{X}_1 - i\hat{Y}_1)$$

$$\left\{ \begin{array}{l} \hat{\alpha}_j^\dagger \hat{\alpha}_k^\dagger = -\hat{\alpha}_k^\dagger \hat{\alpha}_j^\dagger \\ \hat{\alpha}_j \hat{\alpha}_k = -\hat{\alpha}_k \hat{\alpha}_j \end{array} \right. \quad \{ \hat{Z}_k, \hat{X}_k \} = \{ \hat{Z}_k, \hat{Y}_k \} = 0$$

- To restore anti-symmetrization, we add $\hat{Z}$ operators:

$$\hat{\alpha}_k^\dagger = \prod_{j < k} \hat{Z}_j \otimes \frac{1}{2} (\hat{X}_k - i\hat{Y}_k) \otimes \prod_{j > k} \hat{I}_j \quad \hat{\alpha}_k = \prod_{j < k} \hat{Z}_j \otimes \frac{1}{2} (\hat{X}_k + i\hat{Y}_k) \otimes \prod_{j > k} \hat{I}_j$$

Jordan Wigner Mapping

- This is the Jordan Wigner mapping!

$$\hat{\alpha}_k^\dagger = \prod_{j < k} \hat{Z}_j \otimes \frac{1}{2} (\hat{X}_k - i\hat{Y}_k) \otimes \prod_{j > k} \hat{I}_j \quad \hat{\alpha}_k = \prod_{j < k} \hat{Z}_j \otimes \frac{1}{2} (\hat{X}_k + i\hat{Y}_k) \otimes \prod_{j > k} \hat{I}_j$$

- It maps the Hamiltonian of an electronic system to a system of qubits!

$$\hat{H}_{mol} = I^{\otimes 2M} (V_{NN}) + \sum_{p,q} (h_{pq} \hat{\alpha}_p^\dagger \hat{\alpha}_q) + \frac{1}{2} \sum_{p,q,r,s} (h_{pqrs} \hat{\alpha}_p^\dagger \hat{\alpha}_q^\dagger \hat{\alpha}_s \hat{\alpha}_r)$$

- And the electron wavefunctions are represented as state occupations:

$$\vec{s} = \langle s_1, \dots, s_{2M} \rangle \in \{0, 1\}^{2M} \rightarrow \psi_e = \prod_{k=1}^{2M} \left(\hat{\alpha}_k^\dagger \right)^{s_k} |0\dots 0\rangle$$

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Connecting to VQE

- The Jordan Wigner mapping provides the instructions for building a quantum circuit representing the electronic Hamiltonian:

$$\hat{\alpha}_k^\dagger = \prod_{j < k} \hat{Z}_j \otimes \frac{1}{2} (\hat{X}_k - i\hat{Y}_k) \otimes \prod_{j > k} \hat{I}_j \quad \hat{\alpha}_k = \prod_{j < k} \hat{Z}_j \otimes \frac{1}{2} (\hat{X}_k + i\hat{Y}_k) \otimes \prod_{j > k} \hat{I}_j$$

- Non Hartree-Fock ground states can be combined linearly to account for the lost electron correlation and their coefficients can be parametrized, such that the optimal combination can be found.
- This follows, more or less, directly from Adam's talk on VQE.

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